

Features

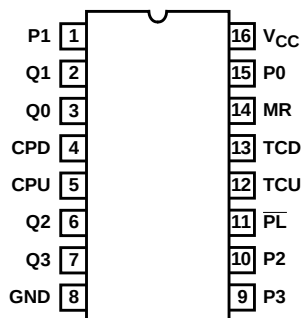
- Synchronous Counting and Asynchronous Loading
- Two Outputs for N-Bit Cascading
- Look-Ahead Carry for High-Speed Counting
- Fanout (Over Temperature Range)
 - Standard Outputs 10 LSTTL Loads
 - Bus Driver Outputs 15 LSTTL Loads
- Wide Operating Temperature Range . . . -55°C to 125°C
- Balanced Propagation Delay and Transition Times
- Significant Power Reduction Compared to LSTTL Logic ICs
- HC Types
 - 2V to 6V Operation
 - High Noise Immunity: $N_{IL} = 30\%$, $N_{IH} = 30\%$ of V_{CC} at $V_{CC} = 5V$
- HCT Types
 - 4.5V to 5.5V Operation
 - Direct LSTTL Input Logic Compatibility, $V_{IL} = 0.8V$ (Max), $V_{IH} = 2V$ (Min)
 - CMOS Input Compatibility, $I_I \leq 1\mu A$ at V_{OL} , V_{OH}

Description

The 'HC192, 'HC193 and 'HCT193 are asynchronously presettable BCD Decade and Binary Up/Down synchronous counters, respectively.

Pinout

CD54HC192, CD54HC193, CD54HCT193 (CERDIP)
CD74HC192 (PDIP, SOP, TSSOP)
CD74HC193 (PDIP, SOIC)
CD74HCT193 (PDIP)
TOP VIEW



Presetting the counter to the number on the preset data inputs (P0-P3) is accomplished by a LOW asynchronous parallel load input ($\overline{PL}$). The counter is incremented on the low-to-high transition of the Clock-Up input (and a high level on the Clock-Down input) and decremented on the low to high transition of the Clock-Down input (and a high level on the Clock-up input). A high level on the MR input overrides any other input to clear the counter to its zero state. The Terminal Count up (carry) goes low half a clock period before the zero count is reached and returns to a high level at the zero count. The Terminal Count Down (borrow) in the count down mode likewise goes low half a clock period before the maximum count (9 in the 192 and 15 in the 193) and returns to high at the maximum count. Cascading is effected by connecting the carry and borrow outputs of a less significant counter to the Clock-Up and Clock-Down inputs, respectively, of the next most significant counter.

If a decade counter is preset to an illegal state or assumes an illegal state when power is applied, it will return to the normal sequence in one count as shown in state diagram.

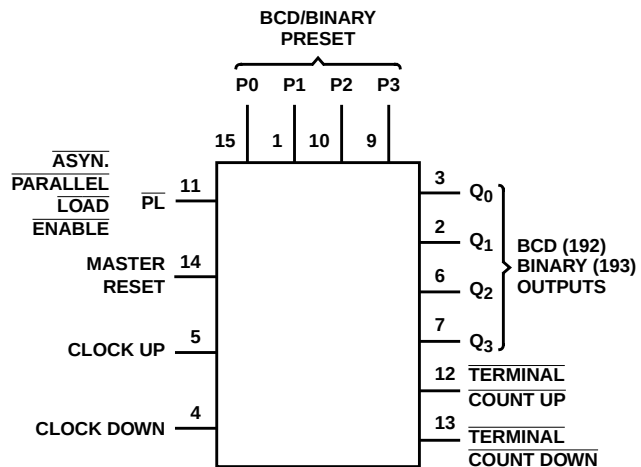
Ordering Information

PART NUMBER	TEMP. RANGE (°C)	PACKAGE
CD54HC192F3A	-55 to 125	16 Ld CERDIP
CD54HC193F3A	-55 to 125	16 Ld CERDIP
CD54HCT193F3A	-55 to 125	16 Ld CERDIP
CD74HC192E	-55 to 125	16 Ld PDIP
CD74HC192NSR	-55 to 125	16 Ld SOP
CD74HC192PW	-55 to 125	16 Ld TSSOP
CD74HC192PWR	-55 to 125	16 Ld TSSOP
CD74HC192PWT	-55 to 125	16 Ld TSSOP
CD74HC193E	-55 to 125	16 Ld PDIP
CD74HC193M	-55 to 125	16 Ld SOIC
CD74HC193MT	-55 to 125	16 Ld SOIC
CD74HC193M96	-55 to 125	16 Ld SOIC
CD74HCT193E	-55 to 125	16 Ld PDIP

NOTE: When ordering, use the entire part number. The suffixes 96 and R denote tape and reel. The suffix T denotes a small-quantity reel of 250.

CD54/74HC192, CD54/74HC193, CD54/74HCT193

Functional Diagram



TRUTH TABLE

CLOCK UP	CLOCK DOWN	RESET	PARALLEL LOAD	FUNCTION
↑	H	L	H	Count Up
H	↑	L	H	Count Down
X	X	H	X	Reset
X	X	L	L	Load Preset Inputs

H = High Voltage Level, L = Low Voltage Level, X = Don't Care, ↑ = Transition from Low to High Level

CD54/74HC192, CD54/74HC193, CD54/74HCT193

Absolute Maximum Ratings

DC Supply Voltage, V_{CC} -0.5V to 7V
 DC Input Diode Current, I_{IK}
 For $V_I < -0.5V$ or $V_I > V_{CC} + 0.5V$ $\pm 20mA$
 DC Output Diode Current, I_{OK}
 For $V_O < -0.5V$ or $V_O > V_{CC} + 0.5V$ $\pm 20mA$
 DC Output Source or Sink Current per Output Pin, I_O
 For $V_O > -0.5V$ or $V_O < V_{CC} + 0.5V$ $\pm 25mA$
 DC V_{CC} or Ground Current, I_{CC} or I_{GND} $\pm 50mA$

Thermal Information

Package Thermal Impedance, θ_{JA} (see Note 1):
 E (PDIP) Package $67^{\circ}C/W$
 M (SOIC) Package $73^{\circ}C/W$
 NS (SOP) Package $64^{\circ}C/W$
 PW (TSSOP) Package $108^{\circ}C/W$
 Maximum Junction Temperature $150^{\circ}C$
 Maximum Storage Temperature Range $-65^{\circ}C$ to $150^{\circ}C$
 Maximum Lead Temperature (Soldering 10s) $300^{\circ}C$
 (SOIC - Lead Tips Only)

Operating Conditions

Temperature Range (T_A) $-55^{\circ}C$ to $125^{\circ}C$
 Supply Voltage Range, V_{CC}
 HC Types 2V to 6V
 HCT Types 4.5V to 5.5V
 DC Input or Output Voltage, V_I , V_O 0V to V_{CC}
 Input Rise and Fall Time
 2V 1000ns (Max)
 4.5V 500ns (Max)
 6V 400ns (Max)

CAUTION: Stresses above those listed in "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress only rating and operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied.

NOTE:

1. The package thermal impedance is calculated in accordance with JESD 51-7.

DC Electrical Specifications

PARAMETER	SYMBOL	TEST CONDITIONS			25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V _I (V)	I _O (mA)	V _{CC} (V)	MIN	TYP	MAX	MIN	MAX	MIN	MAX	
HC TYPES												
High Level Input Voltage	V _{IH}	-	-	2	1.5	-	-	1.5	-	1.5	-	V
				4.5	3.15	-	-	3.15	-	3.15	-	V
				6	4.2	-	-	4.2	-	4.2	-	V
Low Level Input Voltage	V _{IL}	-	-	2	-	-	0.5	-	0.5	-	0.5	V
				4.5	-	-	1.35	-	1.35	-	1.35	V
				6	-	-	1.8	-	1.8	-	1.8	V
High Level Output Voltage CMOS Loads	V _{OH}	V _{IH} or V _{IL}	-0.02	2	1.9	-	-	1.9	-	1.9	-	V
			-0.02	4.5	4.4	-	-	4.4	-	4.4	-	V
			-0.02	6	5.9	-	-	5.9	-	5.9	-	V
High Level Output Voltage TTL Loads			-4	4.5	3.98	-	-	3.84	-	3.7	-	V
			-5.2	6	5.48	-	-	5.34	-	5.2	-	V
Low Level Output Voltage CMOS Loads	V _{OL}	V _{IH} or V _{IL}	0.02	2	-	-	0.1	-	0.1	-	0.1	V
			0.02	4.5	-	-	0.1	-	0.1	-	0.1	V
			0.02	6	-	-	0.1	-	0.1	-	0.1	V
Low Level Output Voltage TTL Loads			4	4.5	-	-	0.26	-	0.33	-	0.4	V
			5.2	6	-	-	0.26	-	0.33	-	0.4	V
Input Leakage Current	I _I	V _{CC} or GND	-	6	-	-	±0.1	-	±1	-	±1	μA
Quiescent Device Current	I _{CC}	V _{CC} or GND	0	6	-	-	8	-	80	-	160	μA

CD54/74HC192, CD54/74HC193, CD54/74HCT193

DC Electrical Specifications (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS			25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
		V _I (V)	I _O (mA)	V _{CC} (V)	MIN	TYP	MAX	MIN	MAX	MIN	MAX	
HCT TYPES												
High Level Input Voltage	V _{IH}	-	-	4.5 to 5.5	2	-	-	2	-	2	-	V
Low Level Input Voltage	V _{IL}	-	-	4.5 to 5.5	-	-	0.8	-	0.8	-	0.8	V
High Level Output Voltage CMOS Loads	V _{OH}	V _{IH} or V _{IL}	-0.02	4.5	4.4	-	-	4.4	-	4.4	-	V
High Level Output Voltage TTL Loads			-4	4.5	3.98	-	-	3.84	-	3.7	-	V
Low Level Output Voltage CMOS Loads	V _{OL}	V _{IH} or V _{IL}	0.02	4.5	-	-	0.1	-	0.1	-	0.1	V
Low Level Output Voltage TTL Loads			4	4.5	-	-	0.26	-	0.33	-	0.4	V
Input Leakage Current	I _I	V _{CC} to GND	-	5.5	-	-	±0.1	-	±1	-	±1	μA
Quiescent Device Current	I _{CC}	V _{CC} or GND	-	5.5	-	-	8	-	80	-	160	μA
Additional Quiescent Device Current Per Input Pin: 1 Unit Load	ΔI _{CC} (Note 2)	V _{CC} - 2.1	-	4.5 to 5.5	-	100	360	-	450	-	490	μA

NOTE:

- For dual-supply systems theoretical worst case (V_I = 2.4V, V_{CC} = 5.5V) specification is 1.8mA.

HCT Input Loading Table

INPUT	UNIT LOADS
P0-P3	0.4
MR	1.45
PL	0.85
CPU, CPD	1.45

NOTE: Unit Load is ΔI_{CC} limit specified in DC Electrical Specifications table, e.g. 360µA max at 25°C.

CD54/74HC192, CD54/74HC193, CD54/74HCT193

Prerequisite For Switching Specifications

PARAMETER	SYMBOL	V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
			MIN	TYP	MAX	MIN	MAX	MIN	MAX	
HC TYPES										
Pulse Width CPU, CPD 192	t _W	2	115	-	-	145	-	175	-	ns
		4.5	23	-	-	29	-	35	-	ns
		6	20	-	-	25	-	30	-	ns
CPU, CPD 193	t _W	2	100	-	-	125	-	150	-	ns
		4.5	20	-	-	25	-	30	-	ns
		6	17	-	-	21	-	26	-	ns
$\overline{\text{PL}}$	t _W	2	80	-	-	100	-	120	-	ns
		4.5	16	-	-	20	-	24	-	ns
		6	14	-	-	17	-	20	-	ns
MR	t _W	2	100	-	-	125	-	150	-	ns
		4.5	20	-	-	25	-	30	-	ns
		6	17	-	-	21	-	26	-	ns
Set-up Time Pn to $\overline{\text{PL}}$	t _{SU}	2	80	-	-	100	-	120	-	ns
		4.5	16	-	-	20	-	24	-	ns
		6	14	-	-	17	-	20	-	ns
Hold Time Pn to $\overline{\text{PL}}$	t _H	2	0	-	-	0	-	0	-	ns
		4.5	0	-	-	0	-	0	-	ns
		6	0	-	-	0	-	0	-	ns
Hold Time CPD to CPU or CPU to CPD	t _H	2	80	-	-	100	-	120	-	ns
		4.5	16	-	-	20	-	24	-	ns
		6	14	-	-	17	-	20	-	ns
Recovery Time $\overline{\text{PL}}$ to CPU, CPD	t _{REC}	2	80	-	-	100	-	120	-	ns
		4.5	16	-	-	20	-	24	-	ns
		6	14	-	-	17	-	20	-	ns
MR to CPU, CPD	t _{REC}	2	5	-	-	5	-	5	-	ns
		4.5	5	-	-	5	-	5	-	ns
		6	5	-	-	5	-	5	-	ns
Maximum Frequency CPU, CPD 192	f _{MAX}	2	5	-	-	4	-	3	-	MHz
		4.5	22	-	-	18	-	15	-	MHz
		6	24	-	-	21	-	18	-	MHz
CPU, CPD 193	f _{MAX}	2	5	-	-	4	-	3	-	MHz
		4.5	25	-	-	20	-	17	-	MHz
		6	29	-	-	24	-	20	-	MHz
HCT TYPES										
Pulse Width CPU, CPD 192	t _W	2	-	-	-	-	-	-	-	ns
		4.5	23	-	-	29	-	35	-	ns
		6	-	-	-	-	-	-	-	ns
CPU, CPD 193	t _W	2	-	-	-	-	-	-	-	ns
		4.5	23	-	-	29	-	35	-	ns
		6	-	-	-	-	-	-	-	ns

CD54/74HC192, CD54/74HC193, CD54/74HCT193

Prerequisite For Switching Specifications (Continued)

PARAMETER	SYMBOL	V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
			MIN	TYP	MAX	MIN	MAX	MIN	MAX	
$\overline{\text{PL}}$	t_W	2	-	-	-	-	-	-	-	ns
		4.5	16	-	-	20	-	24	-	ns
		6	-	-	-	-	-	-	-	ns
MR	t_W	2	-	-	-	-	-	-	-	ns
		4.5	20	-	-	25	-	30	-	ns
		6	-	-	-	-	-	-	-	ns
Set-up Time P _n to $\overline{\text{PL}}$	t_{SU}	2	-	-	-	-	-	-	-	ns
		4.5	15	-	-	19	-	22	-	ns
		6	-	-	-	-	-	-	-	ns
Hold Time P _n to $\overline{\text{PL}}$	t_H	2	-	-	-	-	-	-	-	ns
		4.5	0	-	-	0	-	0	-	ns
		6	-	-	-	-	-	-	-	ns
Hold Time CPD to CPU or CPU to CPD	t_H	2	-	-	-	-	-	-	-	ns
		4.5	16	-	-	20	-	24	-	ns
		6	-	-	-	-	-	-	-	ns
Recovery Time $\overline{\text{PL}}$ to CPU, CPD	t_{REC}	2	-	-	-	-	-	-	-	ns
		4.5	15	-	-	19	-	22	-	ns
		6	-	-	-	-	-	-	-	ns
MR to CPU, CPD	t_{REC}	2	-	-	-	-	-	-	-	ns
		4.5	5	-	-	5	-	5	-	ns
		6	-	-	-	-	-	-	-	ns
Maximum Frequency CPU, CPD 192	f_{MAX}	2	-	-	-	-	-	-	-	MHz
		4.5	22	-	-	18	-	15	-	MHz
		6	-	-	-	-	-	-	-	MHz
CPU, CPD 193	f_{MAX}	2	-	-	-	-	-	-	-	MHz
		4.5	22	-	-	18	-	15	-	MHz
		6	-	-	-	-	-	-	-	MHz

Switching Specifications Input $t_r, t_f = 6\text{ns}$

PARAMETER	SYMBOL	TEST CONDITIONS	V _{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
HC TYPES											
Propagation Delay CPU to $\overline{\text{TCU}}$	$t_{\text{PLH}}, t_{\text{PHL}}$	C _L = 50pF	2	-	-	125	-	155	-	190	ns
		C _L = 50pF	4.5	-	-	25	-	31	-	38	ns
		C _L = 15pF	5	-	10	-	-	-	-	-	ns
		C _L = 50pF	6	-		21	-	26	-	32	ns
CPD to $\overline{\text{TCD}}$	$t_{\text{PLH}}, t_{\text{PHL}}$	C _L = 50pF	2	-	-	125	-	155	-	190	ns
		C _L = 50pF	4.5	-	-	25	-	31	-	38	ns
		C _L = 15pF	5	-	10	-	-	-	-	-	ns
		C _L = 50pF	6	-	-	21	-	26	-	32	ns
CPU to Q _n	$t_{\text{PLH}}, t_{\text{PHL}}$	C _L = 50pF	2	-	-	220	-	270	-	325	ns
		C _L = 50pF	4.5	-	-	43	-	54	-	65	ns
		C _L = 15pF	5	-	18	-	-	-	-	-	ns
		C _L = 50pF	6	-	-	37	-	46	-	55	ns

CD54/74HC192, CD54/74HC193, CD54/74HCT193

Switching Specifications Input $t_r, t_f = 6\text{ns}$ (Continued)

PARAMETER	SYMBOL	TEST CONDITIONS	V_{CC} (V)	25°C			-40°C TO 85°C		-55°C TO 125°C		UNITS
				MIN	TYP	MAX	MIN	MAX	MIN	MAX	
CPD to Q_n	t_{PLH}, t_{PHL}	$C_L = 50\text{pF}$	2	-	-	220	-	270	-	325	ns
		$C_L = 50\text{pF}$	4.5	-	-	43	-	54	-	65	ns
		$C_L = 15\text{pF}$	5	-	18	-	-	-	-	-	ns
		$C_L = 50\text{pF}$	6	-	-	37	-	46	-	55	ns
$\overline{PL}$ to Q_n	t_{PLH}, t_{PHL}	$C_L = 50\text{pF}$	2	-	-	220	-	275	-	330	ns
		$C_L = 50\text{pF}$	4.5	-	-	44	-	55	-	66	ns
		$C_L = 15\text{pF}$	5	-	18	-	-	-	-	-	ns
		$C_L = 50\text{pF}$	6	-	-	37	-	47	-	56	ns
MR to Q_n	t_{PHL}	$C_L = 50\text{pF}$	2	-	-	200	-	250	-	300	ns
		$C_L = 50\text{pF}$	4.5	-	-	40	-	50	-	60	ns
		$C_L = 15\text{pF}$	5	-	17	-	-	-	-	-	ns
		$C_L = 50\text{pF}$	6	-	-	34	-	43	-	51	ns
Transition Time Q, TCU, TCD	t_{TLH}, t_{THL}	$C_L = 50\text{pF}$	2	-	-	75	-	95	-	110	ns
			4.5	-	-	15	-	19	-	22	ns
			6	-	-	13	-	16	-	19	ns
Input Capacitance	C_{IN}	$C_L = 50\text{pF}$	-	-	-	10	-	10	-	10	pF
Power Dissipation Capacitance (Notes 3, 4)	C_{PD}	$C_L = 15\text{pF}$	5	-	40	-	-	-	-	-	pF
HCT TYPES											
Propagation Delay CPU to $\overline{TCU}$	t_{PLH}, t_{PHL}	$C_L = 50\text{pF}$	4.5	-	-	27	-	34	-	41	ns
		$C_L = 15\text{pF}$	5	-	11	-	-	-	-	-	ns
CPU to $\overline{TCD}$	t_{PLH}, t_{PHL}	$C_L = 50\text{pF}$	4.5	-	-	27	-	34	-	41	ns
		$C_L = 15\text{pF}$	5	-	11	-	-	-	-	-	ns
CPU to Q_n	t_{PLH}, t_{PHL}	$C_L = 50\text{pF}$	4.5	-	-	40	-	50	-	60	ns
		$C_L = 15\text{pF}$	5	-	17	-	-	-	-	-	ns
CPD to Q_n	t_{PLH}, t_{PHL}	$C_L = 50\text{pF}$	4.5	-	-	40	-	50	-	60	ns
		$C_L = 15\text{pF}$	5	-	17	-	-	-	-	-	ns
$\overline{PL}$ to Q_n	t_{PLH}, t_{PHL}	$C_L = 50\text{pF}$	4.5	-	-	46	-	58	-	69	ns
		$C_L = 15\text{pF}$	5	-	21	-	-	-	-	-	ns
MR to Q_n	t_{PHL}	$C_L = 50\text{pF}$	4.5	-	-	43	-	54	-	65	ns
		$C_L = 15\text{pF}$	5	-	18	-	-	-	-	-	ns
Transition Time Q, TCU, TCD	t_{TLH}, t_{THL}	$C_L = 50\text{pF}$	2	-	-	75	-	95	-	110	ns
			4.5	-	-	15	-	19	-	22	ns
Input Capacitance	C_{IN}	$C_L = 50\text{pF}$	-	-	-	10	-	10	-	10	pF
Power Dissipation Capacitance (Notes 3, 4)	C_{PD}	$C_L = 15\text{pF}$	5	-	50	-	-	-	-	-	pF

NOTES:

3. C_{PD} is used to determine the dynamic power consumption, per gate.
4. $P_D = V_{CC}^2 f_i + \sum (C_L V_{CC}^2)$ where f_i = Input Frequency, C_L = Output Load Capacitance, V_{CC} = Supply Voltage.

Test Circuits and Waveforms

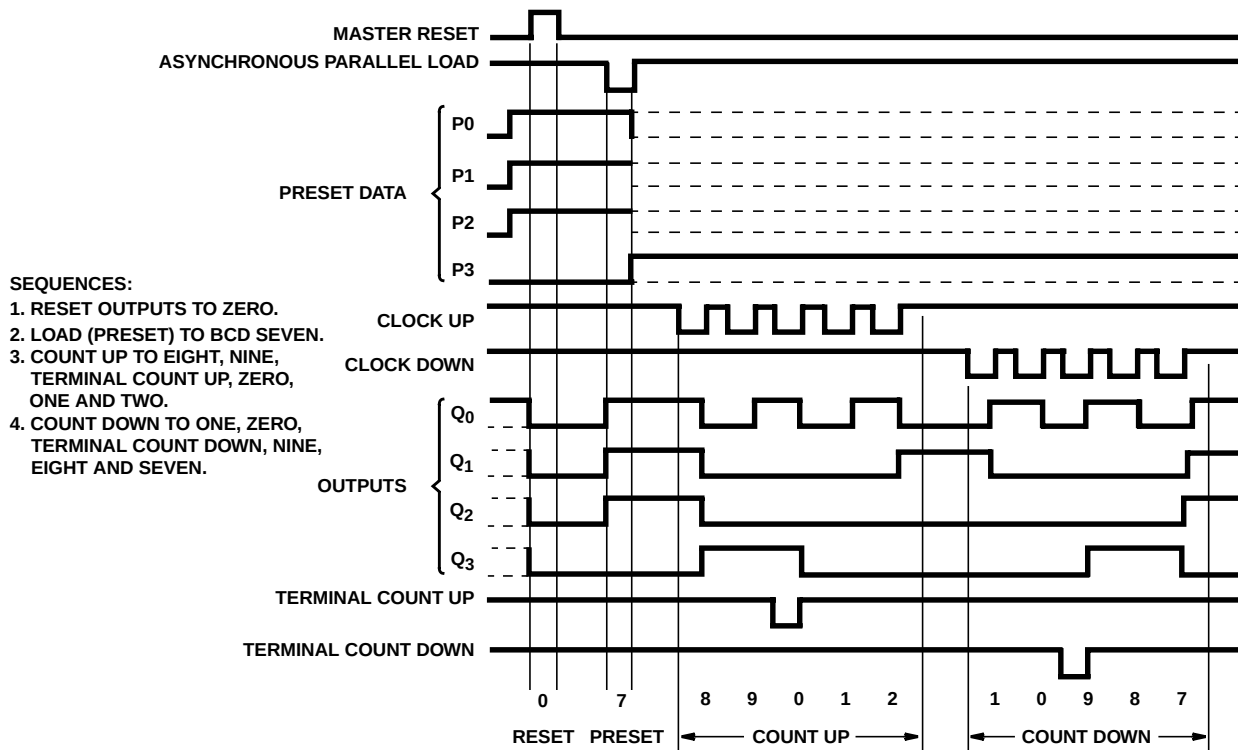


FIGURE 1. 'HC192 SYNCHRONOUS DECADE COUNTERS, TYPICAL RESET, PRESET AND COUNT SEQUENCES

Test Circuits and Waveforms (Continued)

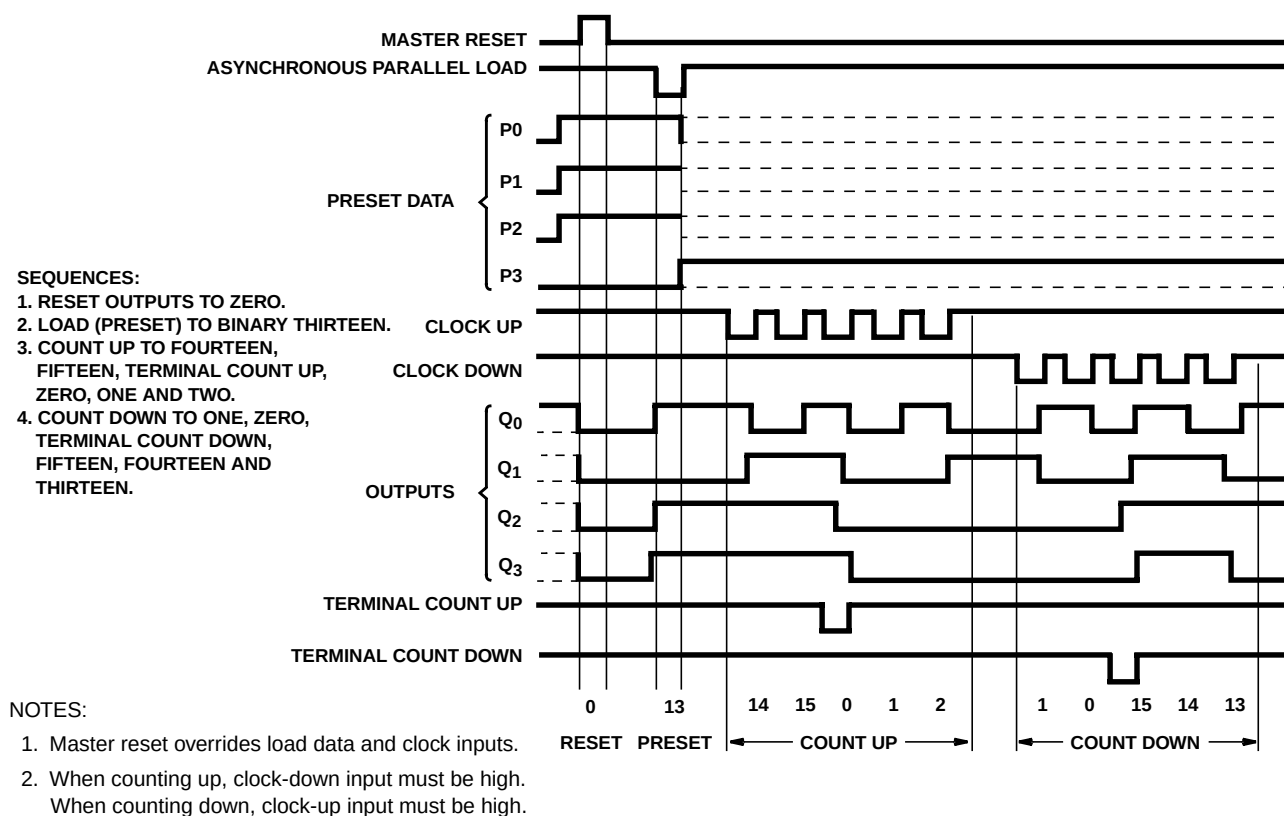


FIGURE 2. 'HC193 SYNCHRONOUS BINARY COUNTERS, TYPICAL RESET, PRESET AND COUNT SEQUENCES

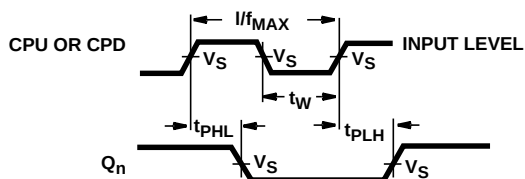


FIGURE 3. CLOCK TO OUTPUT DELAYS AND CLOCK PULSE WIDTH

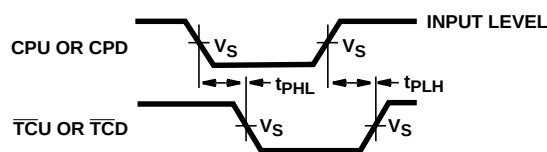


FIGURE 4. CLOCK TO TERMINAL COUNT DELAYS

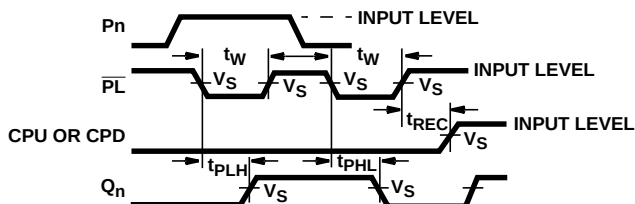


FIGURE 5. PARALLEL LOAD PULSE WIDTH, PARALLEL LOAD TO OUTPUT DELAYS, AND PARALLEL LOAD TO CLOCK RECOVERY TIME

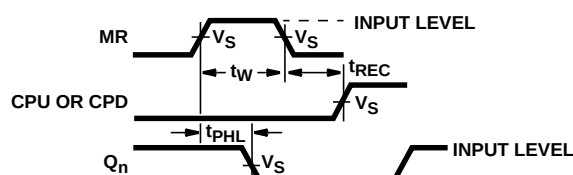


FIGURE 6. MASTER RESET PULSE WIDTH, MASTER RESET TO OUTPUT DELAY AND MASTER RESET TO CLOCK RECOVERY TIME

Test Circuits and Waveforms (Continued)

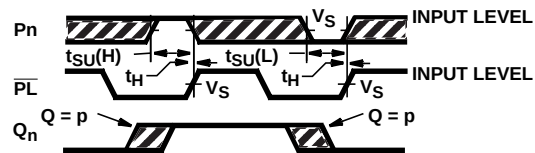


FIGURE 7. SET-UP AND HOLD TIMES DATA TO PARALLEL LOAD (PL)

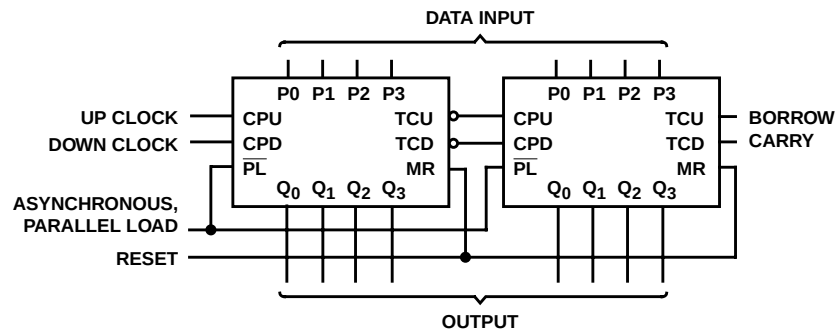
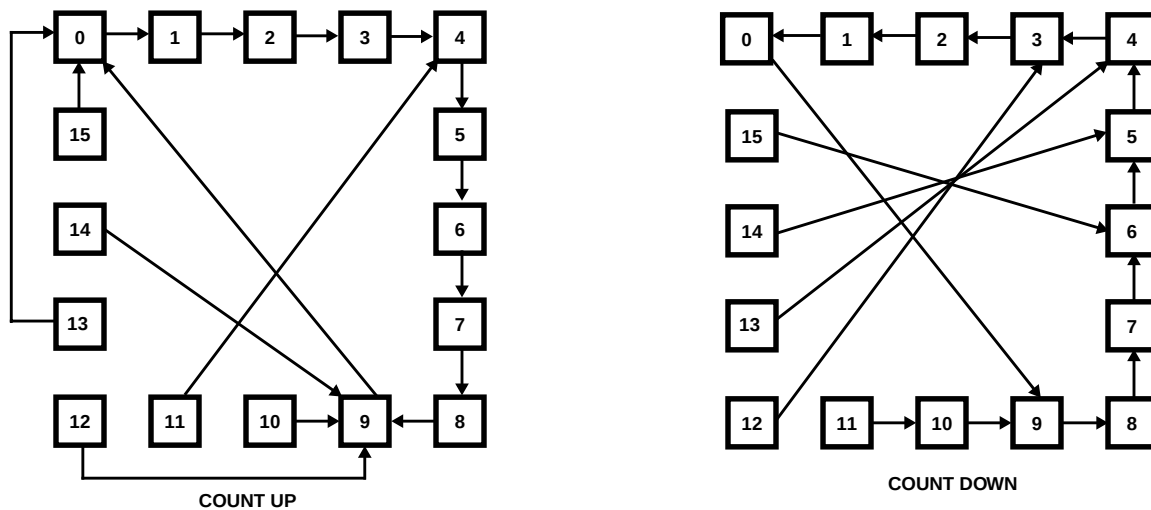


FIGURE 8. CASCADED UP/DOWN COUNTER WITH PARALLEL LOAD



NOTE: Illegal states in BCD counters corrected in one count.

NOTE: Illegal states in BCD counters corrected in one or two counts.

FIGURE 9. 'HC192, 'HCT193 STATE DIAGRAMS

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
5962-8780801EA	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8780801EA CD54HC192F3A
5962-9084801MEA	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-9084801ME A CD54HCT193F3A
CD54HC192F3A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8780801EA CD54HC192F3A
CD54HC192F3A.A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8780801EA CD54HC192F3A
CD54HC193F3A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8772401EA CD54HC193F3A
CD54HC193F3A.A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-8772401EA CD54HC193F3A
CD54HCT193F3A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-9084801ME A CD54HCT193F3A
CD54HCT193F3A.A	Active	Production	CDIP (J) 16	25 TUBE	No	SNPB	N/A for Pkg Type	-55 to 125	5962-9084801ME A CD54HCT193F3A
CD74HC192E	Active	Production	PDIP (N) 16	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC192E
CD74HC192E.A	Active	Production	PDIP (N) 16	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC192E
CD74HC192NSR	Active	Production	SOP (NS) 16	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC192M
CD74HC192NSR.A	Active	Production	SOP (NS) 16	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC192M
CD74HC192NSR.B	Active	Production	SOP (NS) 16	2000 LARGE T&R	-	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC192M
CD74HC192NSRG4	Active	Production	SOP (NS) 16	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC192M
CD74HC192PWR	Active	Production	TSSOP (PW) 16	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HJ192
CD74HC192PWR.A	Active	Production	TSSOP (PW) 16	2000 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HJ192
CD74HC192PWT	Obsolete	Production	TSSOP (PW) 16	-	-	Call TI	Call TI	-55 to 125	HJ192
CD74HC193E	Active	Production	PDIP (N) 16	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC193E
CD74HC193E.A	Active	Production	PDIP (N) 16	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HC193E
CD74HC193M	Obsolete	Production	SOIC (D) 16	-	-	Call TI	Call TI	-55 to 125	HC193M
CD74HC193M96	Active	Production	SOIC (D) 16	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC193M

Orderable part number	Status (1)	Material type (2)	Package Pins	Package qty Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
CD74HC193M96.A	Active	Production	SOIC (D) 16	2500 LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-55 to 125	HC193M
CD74HC193MT	Obsolete	Production	SOIC (D) 16	-	-	Call TI	Call TI	-55 to 125	HC193M
CD74HCT193E	Active	Production	PDIP (N) 16	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HCT193E
CD74HCT193E.A	Active	Production	PDIP (N) 16	25 TUBE	Yes	NIPDAU	N/A for Pkg Type	-55 to 125	CD74HCT193E

⁽¹⁾ **Status:** For more details on status, see our [product life cycle](#).

⁽²⁾ **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

⁽³⁾ **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

⁽⁴⁾ **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

⁽⁵⁾ **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

⁽⁶⁾ **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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OTHER QUALIFIED VERSIONS OF CD54HC192, CD54HC193, CD54HCT193, CD74HC192, CD74HC193, CD74HCT193 :

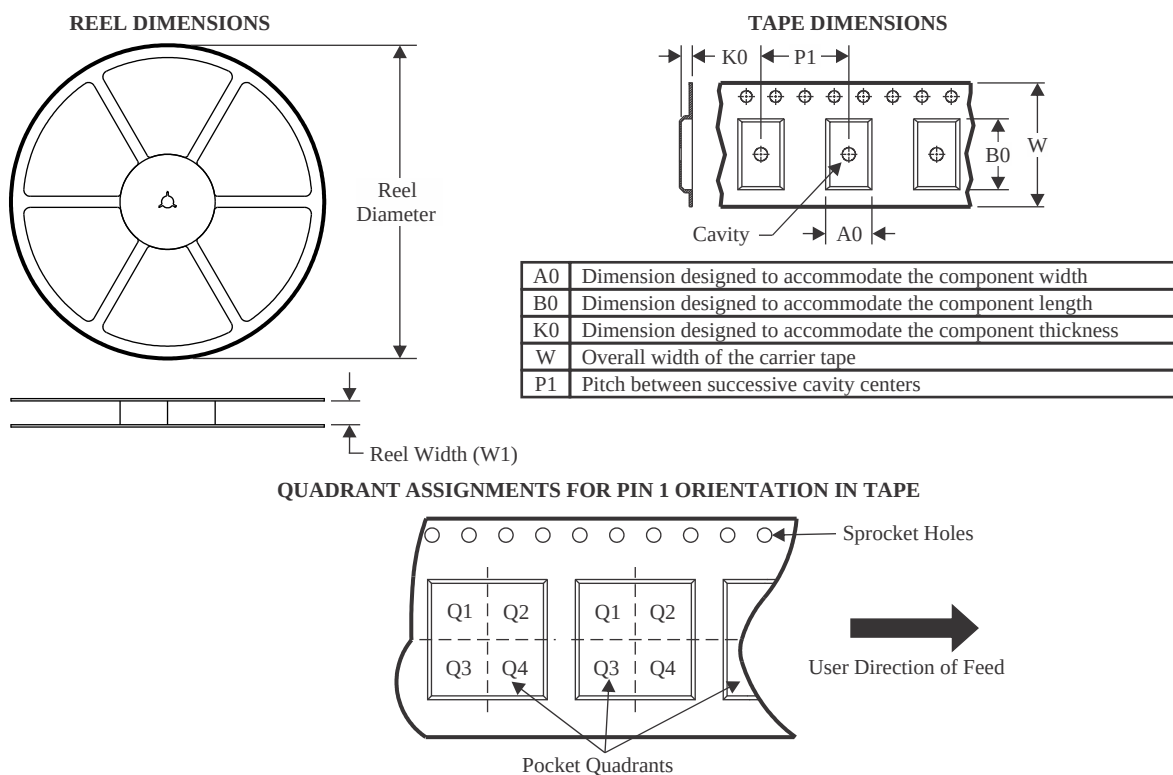
● Catalog : [CD74HC192](#), [CD74HC193](#), [CD74HCT193](#)

- Military : [CD54HC192](#), [CD54HC193](#), [CD54HCT193](#)

NOTE: Qualified Version Definitions:

- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

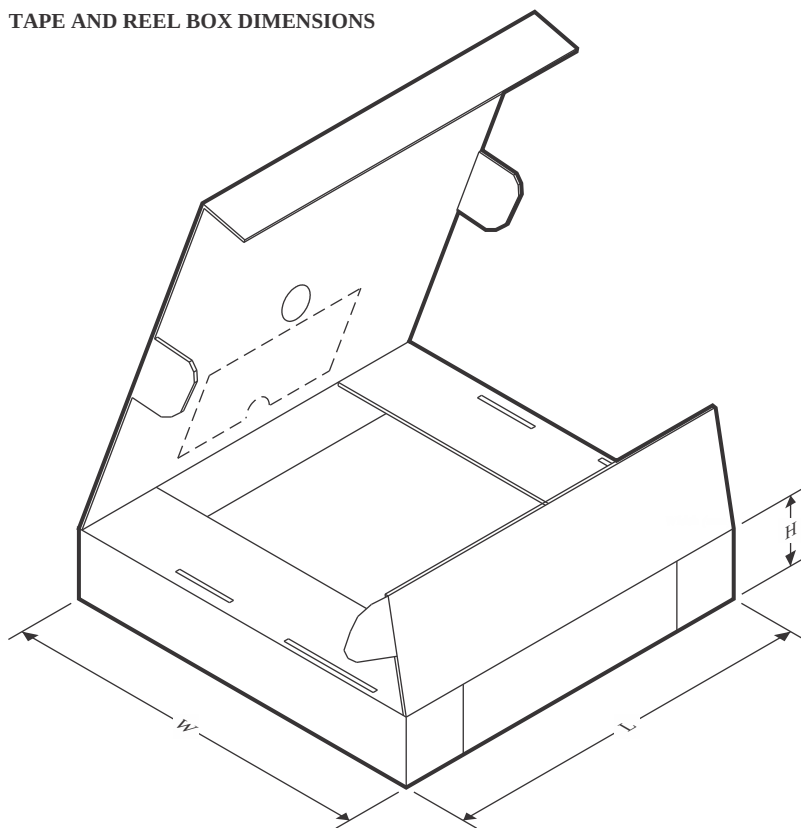
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
CD74HC192NSR	SOP	NS	16	2000	330.0	16.4	8.1	10.4	2.5	12.0	16.0	Q1
CD74HC192PWR	TSSOP	PW	16	2000	330.0	12.4	6.9	5.6	1.6	8.0	12.0	Q1
CD74HC193M96	SOIC	D	16	2500	330.0	16.4	6.5	10.3	2.1	8.0	16.0	Q1

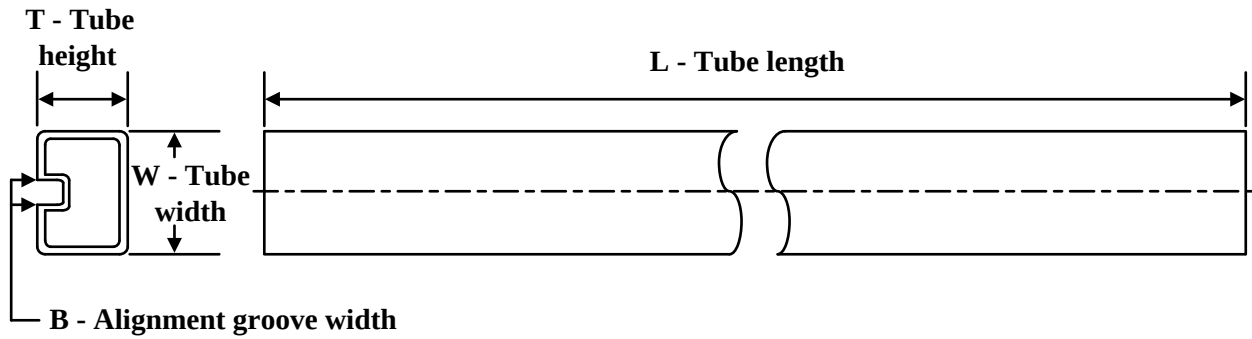
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
CD74HC192NSR	SOP	NS	16	2000	353.0	353.0	32.0
CD74HC192PWR	TSSOP	PW	16	2000	353.0	353.0	32.0
CD74HC193M96	SOIC	D	16	2500	353.0	353.0	32.0

TUBE



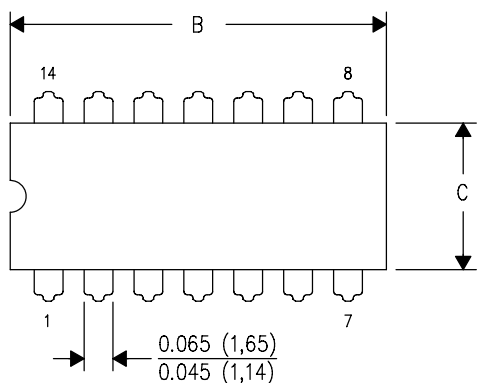
*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
CD74HC192E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC192E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC192E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC192E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC193E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC193E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC193E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HC193E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT193E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT193E	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT193E.A	N	PDIP	16	25	506	13.97	11230	4.32
CD74HCT193E.A	N	PDIP	16	25	506	13.97	11230	4.32

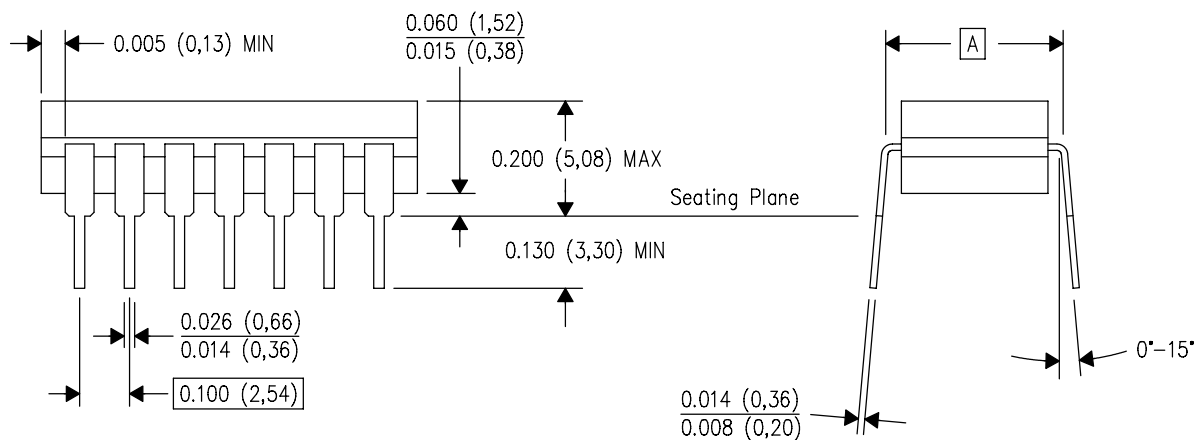
J (R-GDIP-T**)

14 LEADS SHOWN

CERAMIC DUAL IN-LINE PACKAGE

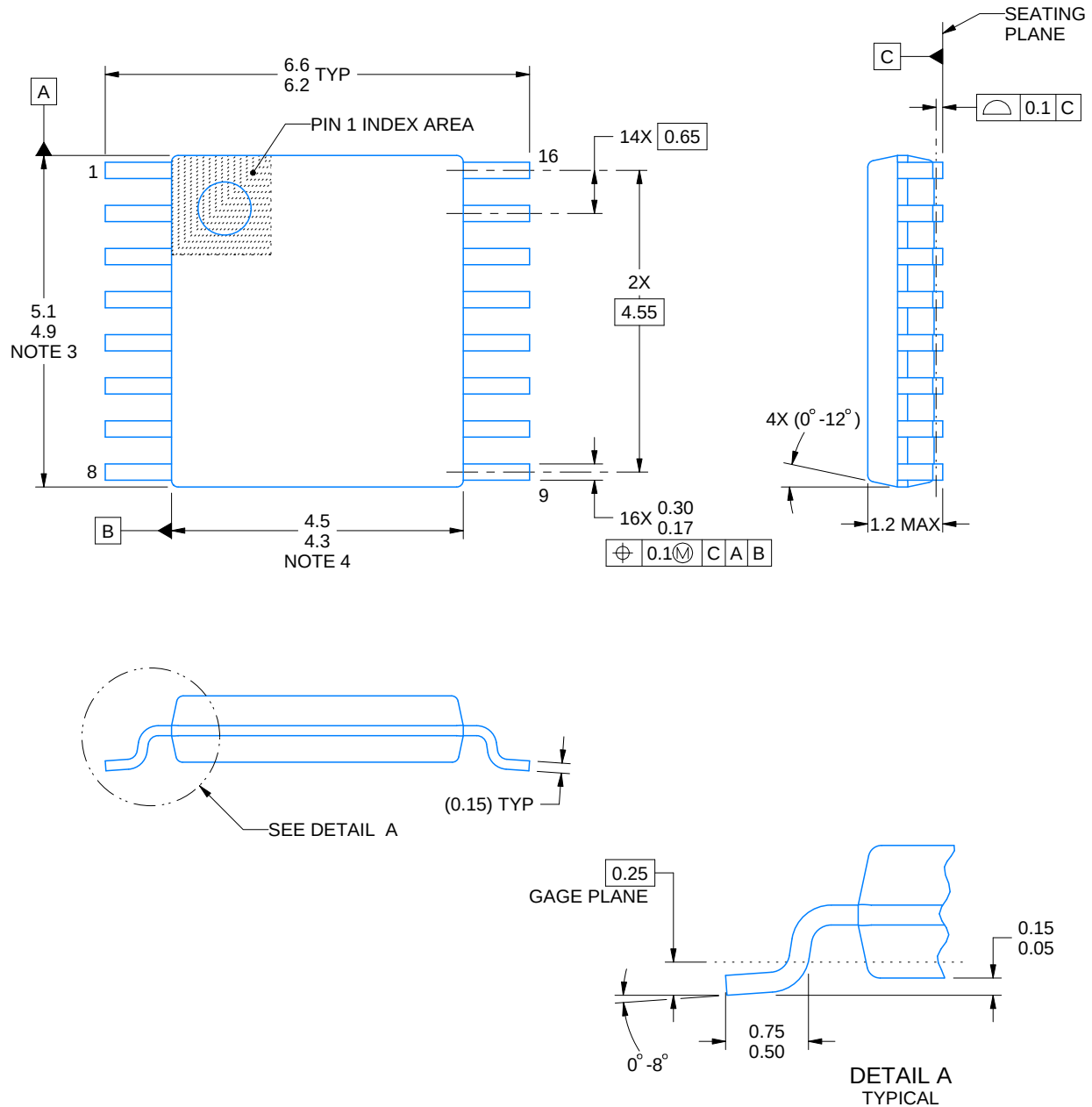
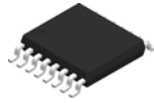


PINS ** DIM	14	16	18	20
A	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC	0.300 (7,62) BSC
B MAX	0.785 (19,94)	.840 (21,34)	0.960 (24,38)	1.060 (26,92)
B MIN	—	—	—	—
C MAX	0.300 (7,62)	0.300 (7,62)	0.310 (7,87)	0.300 (7,62)
C MIN	0.245 (6,22)	0.245 (6,22)	0.220 (5,59)	0.245 (6,22)



4040083/F 03/03

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. This package is hermetically sealed with a ceramic lid using glass frit.
 - D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
 - E. Falls within MIL STD 1835 GDIP1-T14, GDIP1-T16, GDIP1-T18 and GDIP1-T20.



4220204/B 12/2023

NOTES:

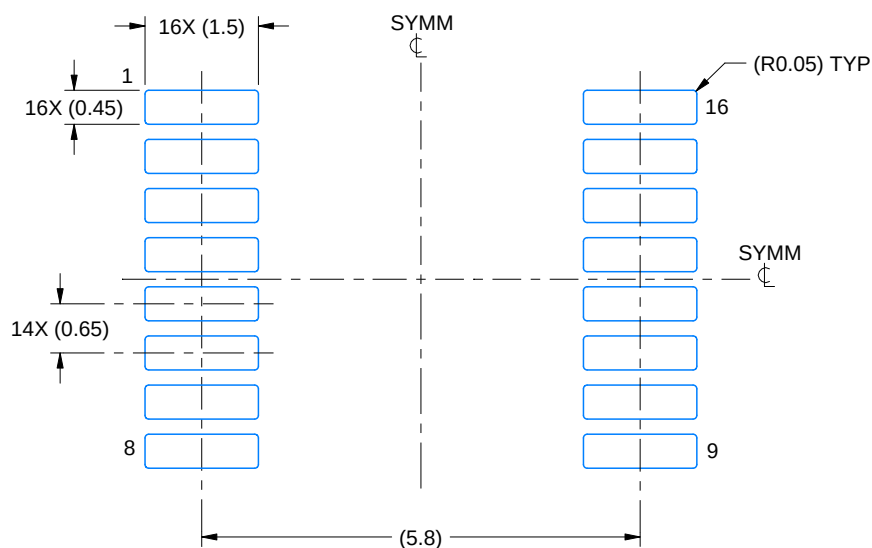
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-153.

EXAMPLE BOARD LAYOUT

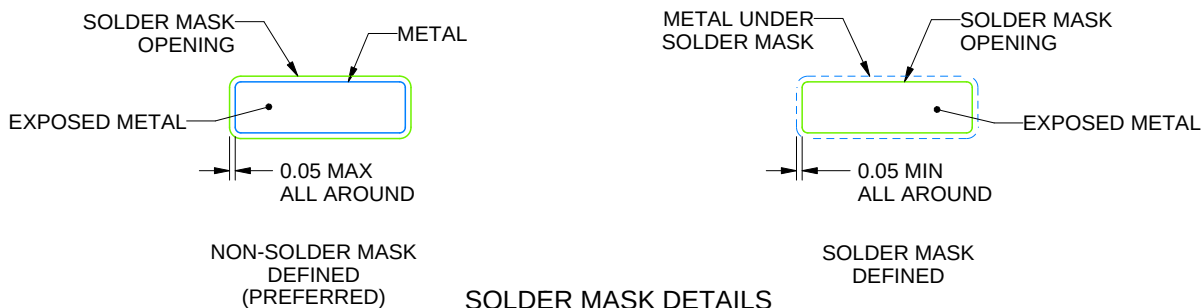
PW0016A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 10X



4220204/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

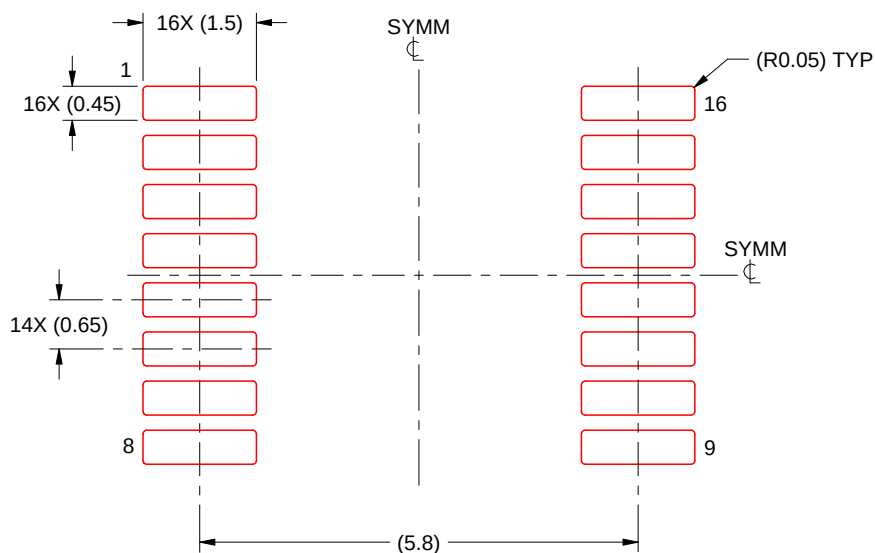
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

PW0016A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE: 10X

4220204/B 12/2023

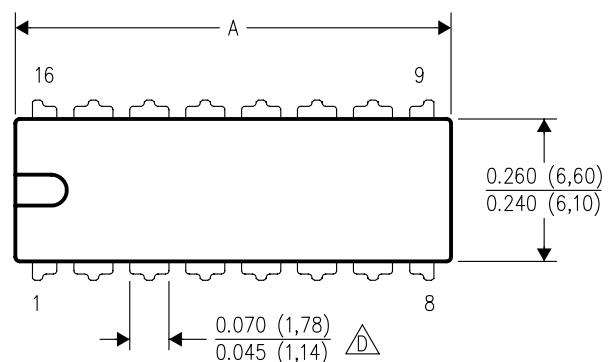
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

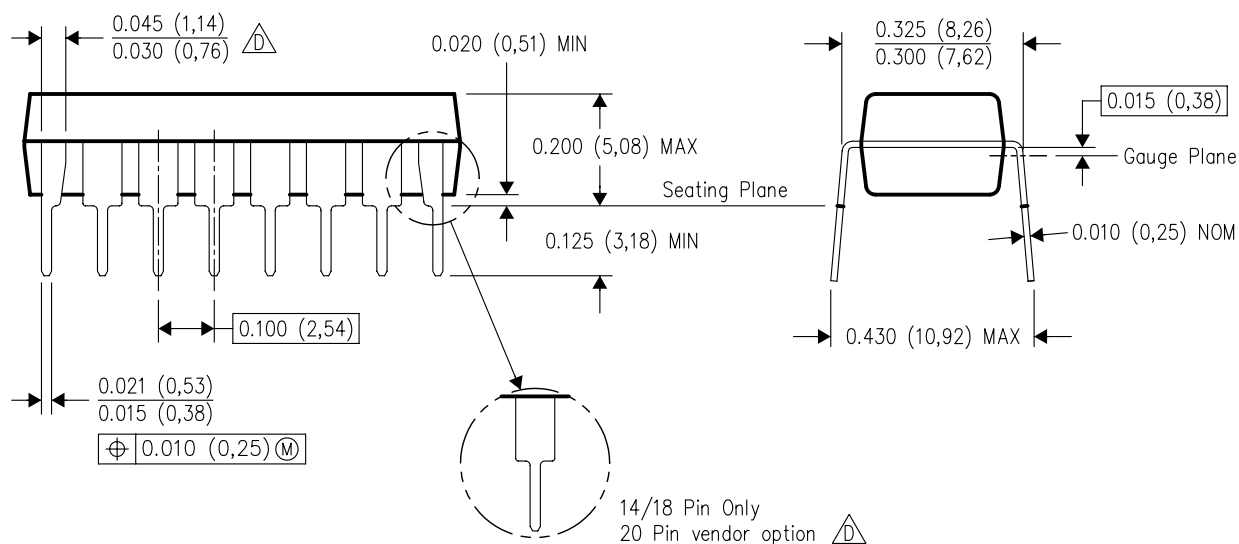
N (R-PDIP-T**)

16 PINS SHOWN

PLASTIC DUAL-IN-LINE PACKAGE



PINS ** DIM	14	16	18	20
A MAX	0.775 (19,69)	0.775 (19,69)	0.920 (23,37)	1.060 (26,92)
A MIN	0.745 (18,92)	0.745 (18,92)	0.850 (21,59)	0.940 (23,88)
MS-001 VARIATION	AA	BB	AC	AD



4040049/E 12/2002

- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 -  Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).
 -  The 20 pin end lead shoulder width is a vendor option, either half or full width.



NS0016A

SOP - 2.00 mm max height

SOP



NOTES:

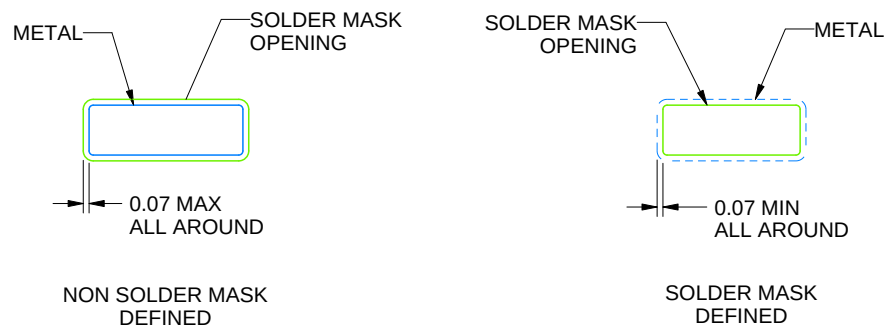
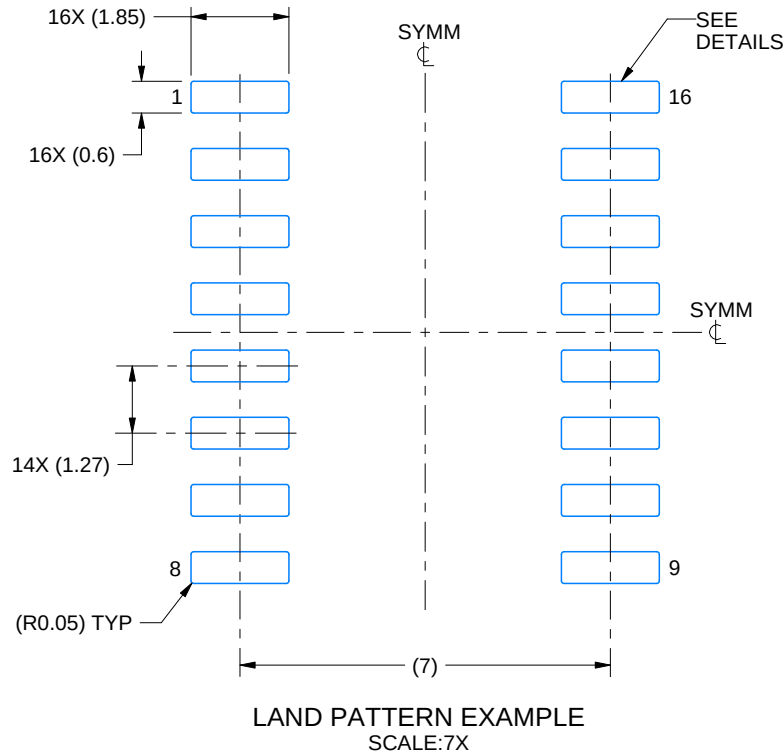
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm, per side.

EXAMPLE BOARD LAYOUT

NS0016A

SOP - 2.00 mm max height

SOP



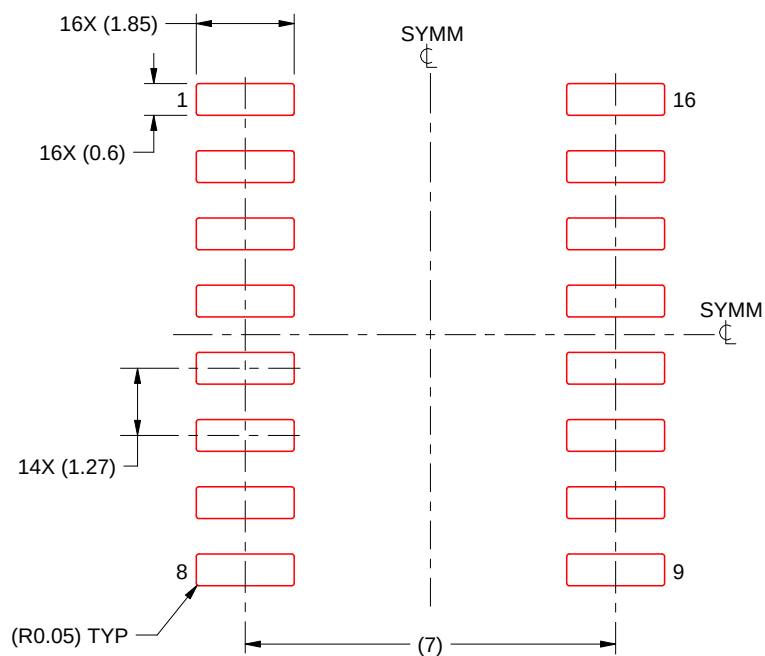
SOLDER MASK DETAILS

4220735/A 12/2021

NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.

6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
 BASED ON 0.125 mm THICK STENCIL
 SCALE:7X

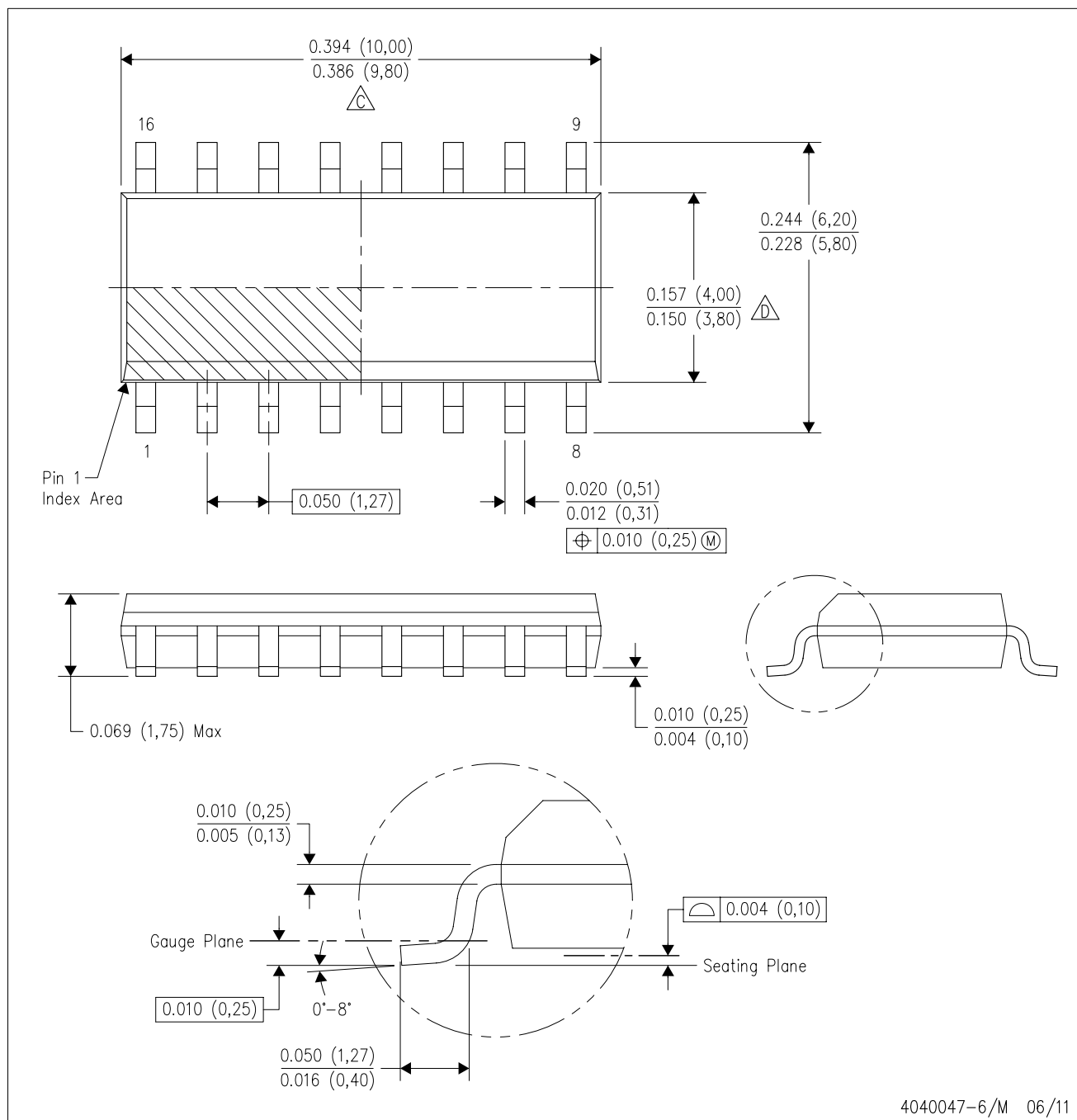
4220735/A 12/2021

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

D (R-PDSO-G16)

PLASTIC SMALL OUTLINE



4040047-6/M 06/11

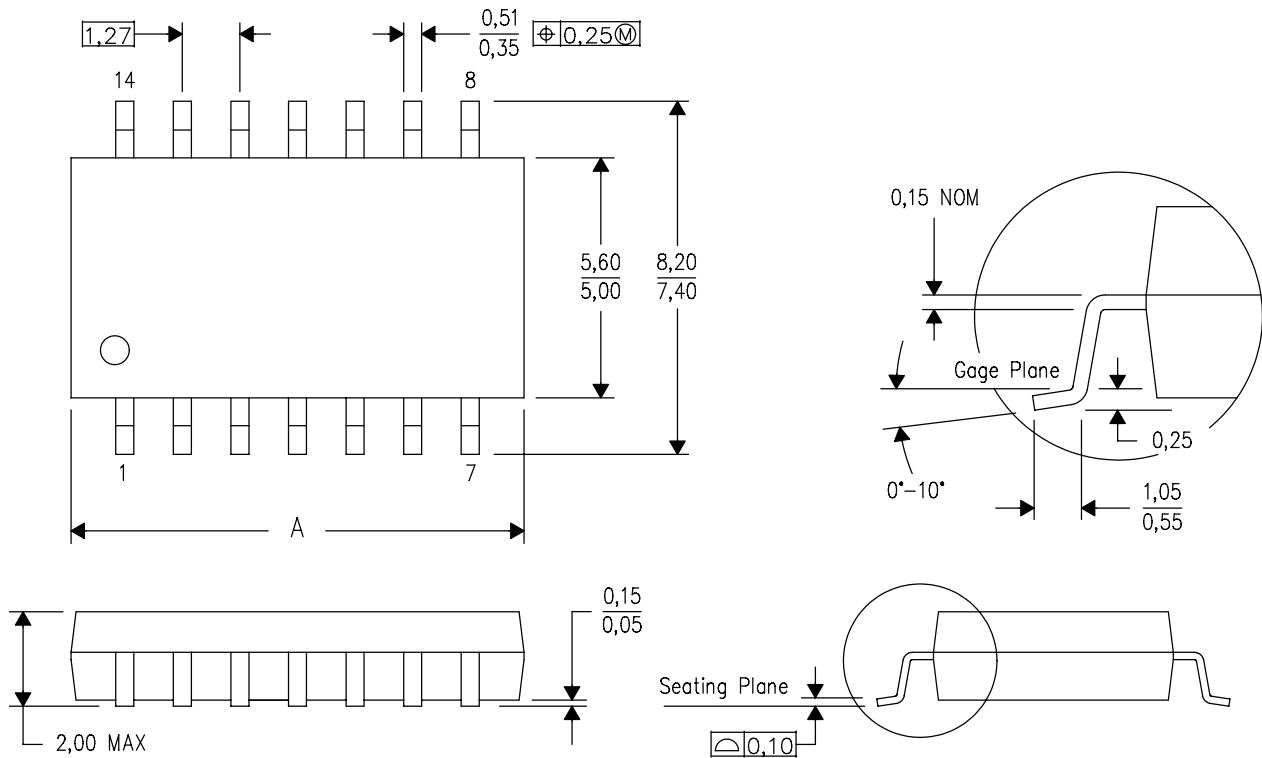
- NOTES:
- A. All linear dimensions are in inches (millimeters).
 - B. This drawing is subject to change without notice.
 - C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.006 (0,15) each side.
 - D. Body width does not include interlead flash. Interlead flash shall not exceed 0.017 (0,43) each side.
 - E. Reference JEDEC MS-012 variation AC.

MECHANICAL DATA

NS (R-PDSO-G**)

PLASTIC SMALL-OUTLINE PACKAGE

14-PINS SHOWN



DIM \ PINS **	14	16	20	24
A MAX	10,50	10,50	12,90	15,30
A MIN	9,90	9,90	12,30	14,70

4040062/C 03/03

- NOTES:
- A. All linear dimensions are in millimeters.
 - B. This drawing is subject to change without notice.
 - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

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